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Original Article

Relationship between pre-procedural non-ischemic ST-segment depression and the clinical outcomes after catheter ablation in persistent atrial fibrillation patients

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ABSTRACT

Background: ST-segment depression suggests the presence of coronary artery disease (CAD) during sinus rhythm, but the clinical significance, including the outcomes after catheter ablation (CA), in atrial fibrillation (AF) patients remain unknown.
Methods: The present study included persistent AF (PerAF) patients from the Osaka Rosai Atrial Fibrillation ablation (ORAF) registry who underwent an initial ablation and had no history of CAD. We assigned the patients based on the presence of ST-segment depression before CA and evaluated the impact of relevant factors on ST-segment depression and the relationship between ST-segment depression, including leads locations (anterior leads, inferior leads, and lateral leads) or depression type (upsloping, horizontal, and downsloping) or the degree of ST-segment depression and late recurrence of AF (LRAF).
Results: This study population included a total of 551 patients of whom 189 had ST-segment depression. The median follow-up duration was 397 days and LRAF occurred in 195 patients. By multiple regression analysis, diabetes mellitus, hemoglobin, brain natriuretic peptide, left ventricular ejection fraction, and left atrial diameter were significant determinants of ST-segment depression before CA. Kaplan-Meier analysis demonstrated that the patients with ST-segment depression had a significantly greater risk of LRAF than those without ($p <$ 0.001). Multivariate Cox proportional hazards analysis showed ST-segment depression was independently and significantly associated with a higher risk of LRAF ($p < 0.001$). The patients with ST-segment depression ≥0.15 mV had a significantly higher risk of LRAF than those with ST-segment depression ≥0.15 mV ($p < 0.001$). No significant differences among the ST-segment depression lead locations and ST-segment depression type were observed.
Conclusion: Non-ischemic ST-segment depression during AF rhythm was significantly associated with LRAF post CA in PerAF patients.

Introduction

Atrial fibrillation (AF) is the most common arrhythmia in the aging society and is associated with considerable morbidity and mortality [1, 2]. Pulmonary vein isolation (PVI) has become established as a standard therapy for patients with drug-refractory paroxysmal atrial fibrillation (PAF) and short-lasting persistent atrial fibrillation (PerAF) [3,4]. Left atrial (LA) dysfunction induces a reduction in total cardiac output and electrocardiogram (ECG) abnormalities are frequently observed in AF patients due to the impact of AF on cardiac function and hemodynamics [5,6]. Previous reports demonstrated that ST-T segment abnormalities and T wave changes were significantly associated with cardiovascular disease mortality [7,8]. There have been studies that have evaluated the predictors of recurrences after PVI [9,10], and various alternations in ECG, including left bundle branch block (LBBB), right bundle branch block (RBBB), and an early repolarization pattern are associated with a high recurrence ratio of AF after PVI [11–13]. Whether there is clinical impact of ECG abnormalities, including ST-segment depression, on the

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outcomes after catheter ablation (CA) in PerAF remains unknown. Therefore, in the present study, we aimed to investigate the relationship between non-ischemic ST-segment depression during AF rhythm before CA and late arrhythmia recurrence in PerAF patients.

Methods

Study population

PerAF patients who underwent an initial ablation were enrolled between May 2014 and September 2020 from the ORAF (Osaka Rosai Atrial Fibrillation ablation) registry [12,14,15]. In the present study, PerAF was defined as AF persisting for >1 week. The patients who had already undergone percutaneous coronary intervention or coronary artery bypass graft, or who had severe stenosis on the coronary angiography or computed tomography before the CA, were excluded.

All patients received a detailed informed consent form and the study protocol was approved by the hospital's institutional review board. The procedure was in accordance with the 'Declaration of Helsinki' and the ethical standards of the responsible committee on human experimentation. This study was granted an exemption from requiring ethics approval by Osaka Rosai Hospital Ethics Committee because this study was a retrospective observational study and the permission for using the clinical data was obtained from all study patients on admission.

Definitions of ECG alternations

The baseline ECGs were provided at the time of admission (1 day before the ablation) and diagnosed by investigating cardiologists in accordance with the American Heart Association/American College of Cardiology/Heart Rhythm Society recommendations [16,17]. LBBB and RBBB were defined as previously described [16]. ST-segment depression was stratified into upsloping, horizontal, and downsloping types (Fig. 1). Inverted T wave included definitely inverted T wave and excluded flat (<10 % of R wave) and low inverted T wave (<−0.1 mV) in the present study [17]. The presence of ST-segment depression or inverted T wave was classified into anterior leads (V1-V4), inferior leads (II, III, aVF), lateral leads (I, aVL, V5, V6), and their combinations (Fig. 1) [18]. The degree of ST-segment depression was defined as the maximal value in the leads with ST-segment depression in the present study.

Echocardiography study

All patients underwent transthoracic echocardiography (TTE) before CA. The TTE was performed with a 2.5 MHz multiplane probe and live images were interpreted by experienced physicians who were blinded to the outcome of the ablation. Comprehensive echocardiographic examinations were performed by trained cardiac sonographers according to the American Society of Echocardiography guidelines [19]. Left ventricular diastolic diameter (LVDd), interventricular septum thickness diameter (IVSTd), and left ventricular posterior wall thickness diameter (LVPWTd) were measured as previously reported. Mean E/e′ was calculated as follows: mean E/e′ = (septal E/e′ + lateral E/e′) / 2 and E/e′ > 9 is defined as one index of LV diastolic dysfunction [20]. Relative wall thickness (RWT) was calculated as follows: RWT = (IVSTd+LVPWTd) / LVDd. Echocardiographic parameters during AF rhythm were acquired where the two preceding cardiac cycles had similar R-R intervals and preferably where the average heart rate was <100 beats per minute. Transesophageal echocardiography prior to the ablation was performed to exclude any LA or LA appendage thrombi.

Ablation procedure

All antiarrhythmic drugs (AADs) were discontinued for at least 3 weeks before the ablation. Anticoagulation therapy was started at least 3 weeks before the ablation procedure. The details of the CA in our laboratory have been published previously [12]. One circular mapping catheter was deployed in the superior and inferior pulmonary veins and the left-sided then right-sided ipsilateral pulmonary veins (PVs) were circumferentially ablated guided by three-dimensional left atrium mapping (CARTO3, Biosense-Webster, Diamond Bar, CA, USA). The PVI was performed with a 3.5 mm ablation catheter with an externally-irrigated tip (ThermoCool® SmartTouch® Catheter, Biosense-Webster). The ablation procedures performed between May 2014 and July 2018 were contact force (CF)-guided, whereas those between August 2018 and September 2020 were ablation index (AI)-guided. After the PVI, the induction of non-PV triggers was performed using isoproterenol and/or adenosine triphosphate. Reproducible non-PV triggers were ablated. Additional ablation, including left atrial posterior wall isolation or superior vena cava isolation, was performed at the operator's discretion. At the end of the procedure, dormant conduction of the PVs was examined with a rapid injection of 40 mg of adenosine

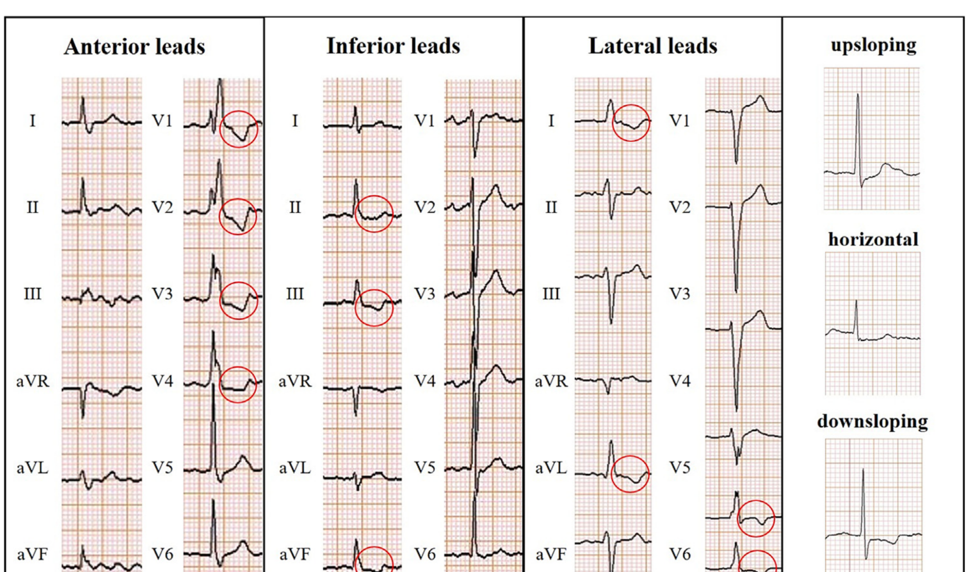

Fig. 1. The lead locations of ST-segment depression (anterior leads, inferior leads, and lateral leads) and ST-segment depression type (upsloping, horizontal, downsloping types).

triphosphate. Additional ablation at the dormant conduction site was performed if necessary.

Follow-up and clinical outcomes

The patients underwent continuous ECG monitoring for approximately 3 days (until discharge) after the ablation. They visited private clinics or our cardiology clinic every 2–4 weeks after the ablation. Patients were encouraged to have smartphone or tablet applications and check their pulse rate and rhythm every day and to visit our hospital if they experienced palpitations or other symptoms. The follow-up visits included a clinical interview, ECG, blood examination, 24-hour Holter monitoring or portable ECG (2-week cardiac event recording), and TTE. Patients with palpitations or other chest symptoms underwent a portable ECG. Late recurrence of AF/atrial tachycardia (LRAF, 3 months after the ablation) was defined as AF/AT documented on the ECG or AF/AT continuing longer than 30 s on the Holter or portable ECG.

The following factors were investigated: 1) The difference in the relationship between ECG abnormalities (ST-segment depression) and incidence of LRAF among the patients with and without ST-segment depression. 2) Relevant factors for pre-procedural ST-segment depression in PerAF patients.

Statistical analysis

JMP 15 statistical software (SAS Institute Inc., Cary, NC, USA) was used for the statistical analysis. Continuous variables were expressed as median [interquartile range]. Normality test was done for continuous variables by Shapiro-Wilk W test. Normal distribution was not confirmed in all variables. Two-group comparisons were analyzed by Mann-Whitney *U* test for continuous variables. Categorical data were expressed as the number (percentage) and were compared using the chi-square test for categorical variables. A multiple regression model to find the relevant parameters for ST-segment depression was performed. Kaplan-Meier curves were used for the incidence of the LRAF and the statistical significance was determined using the log-rank test. A multivariate Cox proportional hazards analysis was performed to compare the hazard ratio of LRAF after CA between the two groups, using the variables including age, gender, hypertension, diabetes mellitus, ST-segment depression, hemoglobin, creatinine, brain natriuretic peptide (BNP), left ventricular (LV) ejection fraction, LA diameter, mean E/e′, β-blocker use, angiotensin-converting enzyme inhibitor (ACE)/angiotensin II receptor blocker (ARB) use, and statin use. Receiver operating characteristic curve (ROC) analysis of the degree of ST-segment depression for predicting LRAF to determine the cut-off values was performed. A value of $p < 0.05$ was considered to be statistically significant in the present study.

Results

Patient and procedural characteristics

This study population included a total of 551 patients from the ORAF registry who underwent an initial ablation using a radiofrequency CA of PerAF and had no history of coronary artery disease (CAD). The clinical characteristics of the enrolled patients are shown in Table 1. Age and BNP were significantly higher, and hemoglobin was significantly lower in the patients with ST-segment depression than in those without ($p = 0.005$, $p < 0.001$, $p < 0.001$, respectively). The ratio of females, chronic heart failure, dyslipidemia and ACEI/ARB use were significantly higher in the patients with ST-segment depression than in those without ($p < 0.001$, $p < 0.001$, $p = 0.043$, and $p = 0.021$, respectively). There was a significant difference in CHADS2 VASc score between the two groups ($p < 0.001$). ECG findings and echocardiographic parameters are shown in Table 2. LBBB and RBBB were confirmed only in the patients with ST-segment depression. The ratio of inverted T wave was significantly higher in the patients with ST-segment depression than in those without ($p < 0.001$). In echocardiographic parameters, IVSTd, LVPWTds, septal E/e′, lateral E/e′, mean E/e′, the ratio of the patients with mean E/e′ > 9, LA diameter, and the ratio of tricuspid regurgitation (TR) were significantly higher, and septal e′, lateral e′, and LA appendage flow were significantly lower in the patients with ST-

Table 1
Baseline patient characteristics.

	Overall population ($n = 551$)	ST-segment depression (+) ($n = 189$)	ST-segment depression (−) ($n = 362$)	*p*-Value
Clinical data				
Age (years)	70 [63, 76]	72 [64, 78]	70 [62, 75]	0.005
Female	173 (31.4 %)	79 (41.8 %)	94 (26.0 %)	<0.001
Body mass index (kg/m^2)	24.2 [21.9, 26.9]	24.0 [21.3, 27.2]	24.3 [22.1, 26.8]	0.747
Hypertension	313 (56.8 %)	116 (61.4 %)	197 (54.4 %)	0.118
Diabetes	94 (17.1 %)	40 (21.2 %)	54 (14.9 %)	0.064
Chronic heart failure	147 (26.7 %)	71 (37.6 %)	76 (21.0 %)	<0.001
Stroke	48 (8.7 %)	21 (11.1 %)	27 (7.5 %)	0.149
Dyslipidemia	135 (24.5 %)	56 (29.6 %)	79 (21.8 %)	0.043
CHADS2 VASc score				<0.001
0	51 (9.3 %)	6 (3.2 %)	45 (12.4 %)	
1	100 (18.1 %)	33 (17.5 %)	67 (18.5 %)	
2	112 (20.3 %)	30 (15.9 %)	82 (22.7 %)	
≥3	288 (52.3 %)	120 (63.5 %)	168 (46.4 %)	
Laboratory data				
Creatinine (mg/dl)	0.9 [0.7, 1.0]	0.8 [0.7, 1.0]	0.9 [0.7, 1.0]	0.525
C-reactive protein (mg/dL)	0.1 [0.1, 0.2]	0.1 [0.1, 0.3]	0.1 [0.1, 0.2]	0.060
Hemoglobin (g/dl)	14.2 [12.9, 15.4]	13.7 [12.4, 15.0]	14.5 [13.2, 15.6]	<0.001
Brain natriuretic peptide (pg/mL)	170.5 [97.2, 274.5]	211.1 [134.8, 320.7]	152.6 [80.9, 254.5]	<0.001
Medications				
DOAC	499 (90.6 %)	167 (88.4 %)	332 (91.7 %)	0.201
AAD	50 (9.1 %)	17 (9.0 %)	33 (9.1 %)	0.962
ACEI/ARB	234 (42.5 %)	93 (49.2 %)	141 (39.0 %)	0.021
Beta-blocker	258 (46.8 %)	95 (50.3 %)	163 (45.0 %)	0.242
Statin	109 (19.8 %)	43 (22.8 %)	66 (18.2 %)	0.206

Continuous data are presented as the median (interquartile range). Categorical variables are presented as numbers (percentage).
AAD, anti-arrhythmic drug; ACEI, angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blocker; DOAC, direct oral anticoagulant.

Table 2
Electrocardiographic findings and echocardiographic parameters.

	Overall population (n = 551)	ST-segment depression (+) (n = 189)	ST-segment depression (−) (n = 362)	p-Value
Electrocardiographic findings				
Complete right bundle branch block	2 (0.4 %)	2 (1.1 %)	0 (0 %)	0.050
Complete left bundle branch block	35 (6.4 %)	35 (18.5 %)	0 (0 %)	<0.001
ST segment depression	189 (34.3 %)	189 (100 %)	–	
Anterior leads	61 (11.1 %)	61 (32.2 %)	–	
Inferior leads	81 (14.7 %)	81 (42.9 %)	–	
Lateral leads	135 (24.5 %)	135 (71.4 %)	–	
Inverted T wave	172 (31.2 %)	113 (59.8 %)	59 (16.3 %)	<0.001
Anterior leads	80 (14.5 %)	54 (28.6 %)	26 (7.2 %)	<0.001
Inferior leads	90 (16.3 %)	60 (31.7 %)	30 (8.3 %)	<0.001
Lateral leads	81 (14.7 %)	63 (33.3 %)	18 (5.0 %)	<0.001
Heart rate (bpm)	89 [76, 103]	89 [78, 106]	89 [75, 102]	0.382
Echocardiographic parameters				
LVDd (mm)	49 [46, 52]	49 [45, 53]	49 [46, 51]	0.361
LVDs (mm)	31 [28, 35]	31 [28, 37]	31 [28, 34]	0.140
LVEF (%)	65 [59, 70]	64 [55, 69]	65 [60, 70]	0.061
IVSTd (mm)	9 [9, 10]	9 [9, 10]	9 [8, 10]	0.001
LVPWTd (mm)	9 [9, 10]	9 [9, 10]	9 [8, 10]	0.010
RWT	0.38 [0.35, 0.40]	0.38 [0.35, 0.41]	0.37 [0.35, 0.40]	0.078
Peak E velocity	90 [76, 104]	90 [75, 104]	90 [77, 104]	0.903
Septal e′	8.2 [6.7, 9.9]	7.2 [5.9, 9.3]	8.6 [7.1, 10.3]	<0.001
Lateral e′	10.9 [9.0, 12.9]	10.3 [8.5, 12.2]	11.2 [9.5, 13.3]	<0.001
Septal E/e′	11.0 [8.9, 13.9]	12.2 [10.1, 15.0]	10.2 [8.4, 13.2]	<0.001
Lateral E/e′	8.3 [6.7, 10.2]	8.7 [7.0, 11.5]	7.9 [6.4, 10.0]	0.002
Mean E/e′	9.6 [7.8, 12]	10.4 [8.7, 13.0]	9.3 [7.5, 11.5]	<0.001
Mean E/e′ > 9	324 (58.8)	131 (69.3)	193 (59.3)	<0.001
LA diameter (mm)	47 [43, 51]	49 [44, 53]	47 [43, 50]	<0.001
Mitral regurgitation ≥III	31 (5.6 %)	12 (6.3 %)	19 (5.2 %)	0.595
Tricuspid regurgitation ≥III	55 (10.0 %)	27 (14.3 %)	28 (7.7 %)	0.014
LAAV (m/s)	32 [23, 43]	29 [21, 40]	34 [25, 46]	<0.001

Continuous data are presented as the median (interquartile range). Categorical variables are presented as numbers (percentage).
IVSTd, intraventricular septal thickness at end-diastole; LAAV, left atrial appendage flow; LA, left atrium; LVDd, left ventricular end-diastolic diameter; LVDs, left ventricular end-systolic diameter; LVEF, left ventricular ejection fraction; LVPWTd, left ventricular posterior wall thickness at end-diastole; RWT, relative wall thickness.

segment depression than in those without. The procedural characteristics of the patients are shown in Table 3.

Relationship between ST-segment depression and relevant factors in PerAF patients

We investigated the relationship between ST-segment depression in PerAF patients before CA and relevant parameters by multiple regression analysis. Diabetes mellitus, hemoglobin, BNP, LV ejection fraction, and LA diameter were independent and significant determinants of ST-segment depression in PerAF patients before CA (Table 4).

Clinical outcomes associated with ECG abnormalities

The median follow-up duration was 397 [183, 1001] days and LRAF occurred in 195 patients (35.4 %). Kaplan-Meier analysis demonstrated that the patients with ST-segment depression had a significantly greater risk of LRAF than those without ($p < 0.001$) (Fig. 2A). In the present study, RBBB, LBBB (QRS duration>120 msec), and pacing rhythm comprised 35, 2, and

Table 3
Procedural characteristics.

	Overall population (n = 551)	ST-segment depression (+) (n = 189)	ST-segment depression (−) (n = 362)	p-Value
Procedural characteristics				
Procedure time (min)	165 [130, 215]	165 [135, 215]	165 [130, 215]	0.728
Number of applications	92 [75, 116]	92 [75, 115]	93 [73, 119]	0.595
Total time of the applications (min)	38 [29, 48]	37 [29, 46]	38 [29, 50]	0.784
Fluoroscopy duration (min)	22 [13, 41]	23 [13, 43]	21 [13, 40]	0.317
CTIB	306 (55.5 %)	91 (48.1 %)	215 (59.4 %)	0.012
SVCI	46 (8.3 %)	16 (8.5 %)	30 (8.3 %)	0.943
LAPWI	78 (14.2 %)	26 (13.8 %)	52 (14.4 %)	0.846
Other non-PV foci ablation	23 (4.2 %)	7 (3.7 %)	16 (4.4 %)	0.690
Complications				
Cardiac tamponade	0 (0 %)	0 (0 %)	0 (0 %)	–
Pseudo aneurysm	2 (0.4 %)	0 (0 %)	2 (0.6 %)	0.306
Phrenic nerve palsy	4 (0.7 %)	2 (1.1 %)	2 (0.6 %)	0.507
Bleeding at the puncture site	0 (0 %)	0 (0 %)	0 (0 %)	–
Gastroparesis	2 (0.4 %)	0 (0 %)	2 (0.6 %)	0.306
Embolism	0 (0 %)	0 (0 %)	0 (0 %)	–

Continuous data are presented as the median (interquartile range). Categorical variables are presented as numbers (percentage).
CTIB, cavo tricuspid isthmus block; LAPWI, left atrial posterior wall isolation; PV, pulmonary vein; SVCI, superior vena cava isolation.

Table 4
Multivariate logistic regression analysis for ST-segment depression.

	OR	95 % CI	*p*-Value
Age	1.003	0.980–1.026	0.818
Female	1.547	0.951–2.519	0.079
Body mass index	1.020	0.965–1.078	0.475
Hypertension	1.072	0.704–1.633	0.745
Diabetes mellitus	1.773	1.069–2.942	0.027
Dyslipidemia	1.125	0.713–1.773	0.613
Heart rate	0.993	0.983–1.002	0.124
Hemoglobin	0.874	0.770–0.991	0.036
Creatinine	0.702	0.420–0.941	0.073
Log BNP	2.188	1.153–4.235	0.018
LVEF	0.969	0.951–0.988	0.002
LA diameter	1.052	1.012–1.095	0.010
Mean E/e′	1.009	0.951–1.070	0.773

BNP, brain natriuretic peptide; CI, confidence interval; LA, left atrium; LVEF, left ventricular ejection fraction; OR, odds ratio.

2 patients, respectively, and those patients had ST-segment depression (Table 2). Kaplan-Meier analysis, excluding the patients with bundle branch block and pacing rhythm, showed that the patients with ST-segment depression had a significantly higher risk of LRAF than those without, which was the same as overall populations (Fig. 2B). Kaplan-Meier analyses according to the lead locations of ST-segment depression (anterior leads, inferior leads, and lateral leads and their combinations) and ST-segment depression type (upsloping, horizontal, downsloping, and their combination types) are shown in Fig. 3. No significant differences among the patients stratified by the lead locations of ST-segment depression or between the patients with single lead (only anterior leads, inferior leads, or lateral leads) and multiple leads were observed (Fig. 3A and B, respectively). Among the patients with ST-segment depression type (upsloping, horizontal, downsloping, and their combination type), there were no significant differences in the LRAF (Fig. 3C). We used ROC curve analyses of the degree of ST-segment depression for predicting LRAF to determine the cut-off value. The optimal cut-off values for predicting LRAF was the degree of ST-segment depression = 0.15 mV (area under the curve = 0.74, sensitivity = 55.7 %, specificity = 80.4 %) (Online Fig. 1). The patients with ST-segment depression ≥0.15 mV had a significantly higher risk of LRAF than those with ST-segment depression ≥0.15 mV ($p < 0.001$) (Fig. 3D). Multivariate Cox proportional hazards analysis showed that ST-segment depression was independently and significantly associated with a higher risk of the LRAF [hazard ratio (HR): 2.314; 95 % confidence interval (CI): 1.716–3.122, $p < 0.001$], in addition to hemoglobin (HR: 0.907; 95 % CI: 0.824–0.999, $p = 0.046$) and LA diameter (HR: 1.032; 95 % CI: 1.003–1.061, $p = 0.028$) (Fig. 4).

Discussion

Main findings

The main findings in the present study were that (1) the PerAF patients with non-ischemic ST-segment depression had a significantly greater risk of LRAF than those without ST-segment depression after CA ($p < 0.001$). (2) The patients with ST-segment depression ≥0.15 mV had a significantly higher risk of LRAF than those with ST-segment depression ≥0.15 mV, while the lead locations of ST-segment depression and ST-segment depression type were not related to LARF after CA. These results suggested that ST-segment depression during AF rhythm was a useful predictor of the poor outcome after CA in PerAF patients who had no history of CAD.

ST-segment depression during AF rhythm

The prevalence of ST-segment depression in the PerAF patients who underwent CA was 35.6 % in the present study. ST-segment depression suggested the presence of CAD during the exercise stress test in sinus rhythm, but previous reports have shown that the presence of CAD in AF patients with ST-segment depression is only 32.5 % [21]. Another study showed that the sensitivity and specificity of ST-segment depression for the presence of CAD are only 35 % and 75 %, respectively in AF patients with rapid heart rate [22]. ST-segment depression can be an indicator of the presence of CAD, but the patients who had history of CAD were excluded from our study. The mechanism of non-ischemic ST-segment depression in AF patients remains unknown. Electrolyte shifts and sympathetic neurogenic stimulation at the cellular level may play an important role in non-ischemic ST-segment depression [23]. Exercise-induced ST-segment depression in patients with a low likelihood of ischemic heart disease is related to increased β-adrenergic sensitivity regarding chronotropic and electrophysiological, but not inotropic responses [24]. The diastolic dysfunction can be an early sign of myocardial ischemia and other metabolic disturbances, which appear before any contraction abnormality can be detected [25,26]. These changes may be linked to microvascular disease and/or endothelial dysfunction and raise the possibility that microvascular abnormalities might contribute to the ST-segment depression [27–29]. In the present study, IVSTd, LVPWTds, E/e′, and LA diameter were significantly higher, and e′ was significantly lower in the patients with ST-segment depression than in those without. LV diastolic dysfunction and LA dysfunction might induce sympathetic activation, microvascular disease, and endothelial dysfunction, and these structural abnormalities might have affected ST-segment depression in the enrolled patients.

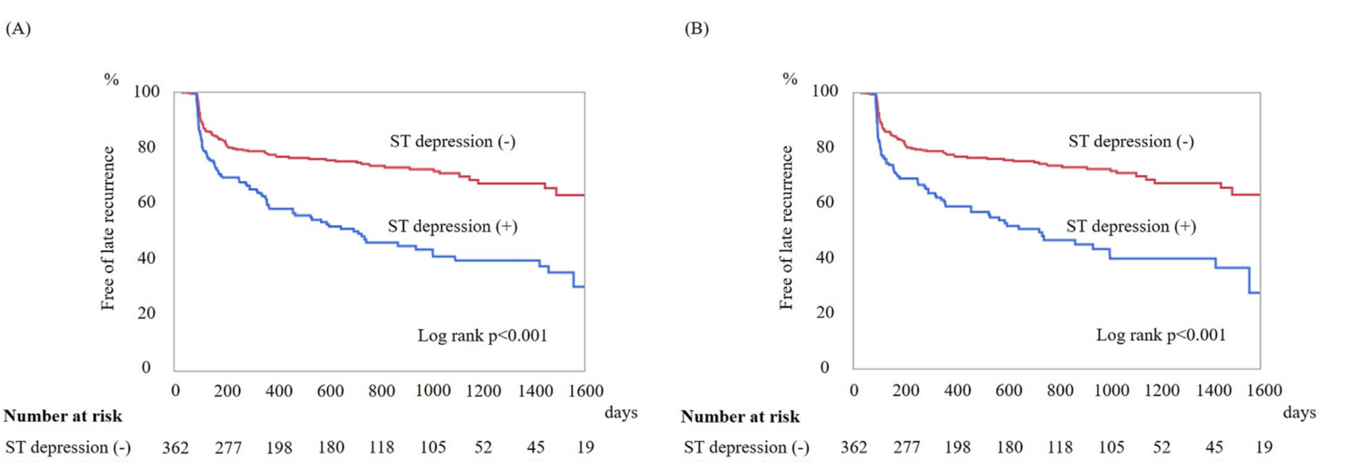

Fig. 2. (A) Kaplan-Meier curve of the LRAF in patients with and without ST-T segment depression. (B) Kaplan-Meier curve of the LRAF in patients with and without ST-T segment depression, excluding those with QRS duration >120 msec (right bundle branch block, left bundle branch block, and pacing rhythm). LRAF, late recurrence of atrial fibrillation/atrial tachycardia.

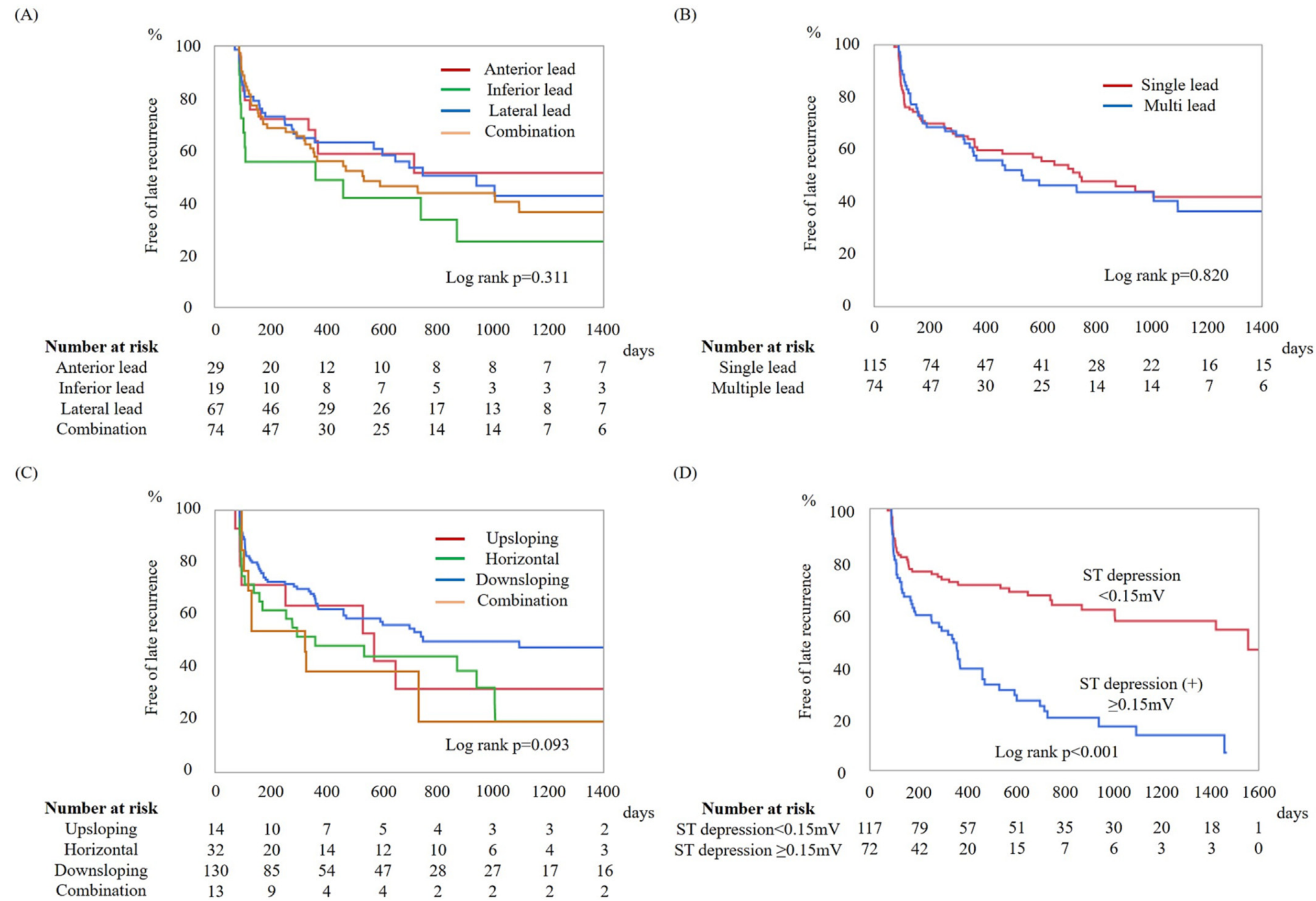

Fig. 3. (A) Kaplan-Meier curves of the LRAF among the patients stratified by the location of ST-segment depression. (B) Kaplan-Meier curves of the LRAF between the patients with between single lead (only anterior leads, inferior leads, lateral leads) and multiple leads. (C) Kaplan-Meier curves of the LRAF among the patients according to ST-segment depression type (upsloping, horizontal, downsloping, and their combination types). (D) Kaplan-Meier curves of the LRAF among the patients according to the degree of ST-segment depression (<0.15 mV and ≥0.15 mV). LRAF, late recurrence of atrial fibrillation/atrial tachycardia.

ST-segment depression and late arrhythmia recurrence after CA

Multivariate Cox proportional hazard analysis showed that ST-segment depression was significantly associated with a higher risk of LRAF (Fig. 4). Non-ischemic ST-segment depression suggests the presence of cardiac structural abnormalities, including LV systolic and/or diastolic dysfunction and LA dysfunction. These alternations are associated with AF initiation and perpetuation due to abnormal Ca^{2+} handling and abnormalities of the atrial cardiomyocyte, fibrotic changes, and alterations in the interstitial matrix with primarily non-collagen deposits [30]. In patients with non-ST segment elevation myocardial infarction, degree of ST-segment depression on admission ECG predicts in-hospital mortality [31] and the report suggests that there is a significant relationship between the degree of ST-segment depression and cardiac structural abnormalities.

We found no significant difference in the LRAF according to the location of ST-segment depression. Previous reports showed that ST-segment depression in the lateral leads on hospital admission predicts a poor in-hospital outcome in patients with a first non-ST segment elevation acute myocardial infarction and ST-segment depression in the

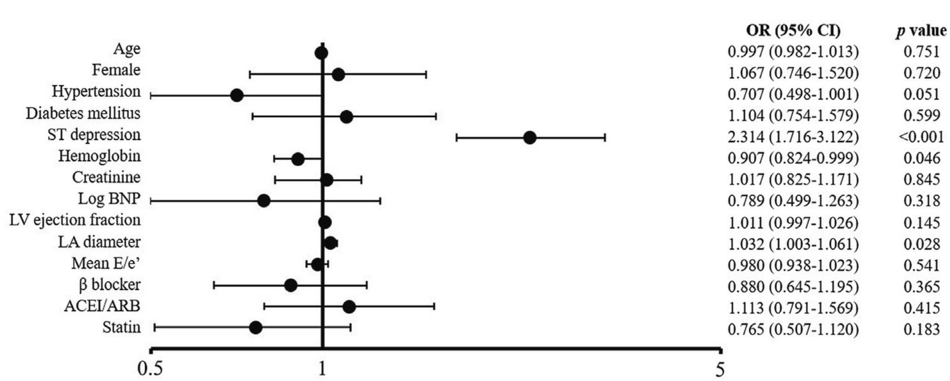

Fig. 4. Multivariate Cox proportional hazard model for LRAF. The results are illustrated as Forest plot. ACEI, angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blocker; BNP, brain natriuretic peptide; CI, confidence interval; LRAF, late recurrence of atrial fibrillation/atrial tachycardia; LA, left atrium; LV, left ventricle; OR, odds ratio.

high lateral leads could be a prognostic factor in hypertrophic cardiomyopathy patients [32,33]. ST-segment depression in lateral leads is thought to be associated with more extensive CAD in acute myocardial infarction patients with non-ST segment elevation and reflects a severe reduction in the myocardial blood flow or other pathophysiologic substrates, including extensive disarray or fibrosis in hypertrophic cardiomyopathy patients [32,33]. In the present study, ST-segment depression in lateral leads comprised 150 patients (71.4 %), including the patients with ST-segment depression in multiple leads. LA remodeling progresses in PerAF patients and LA dysfunction may contribute significantly, up to one-third, to total cardiac output [5]. The high prevalence of ST-segment depression in lateral leads in PerAF patients might be due to a reduction in the whole myocardial blood flow. A previous study demonstrated that the location of inverted T wave is not associated with cardiac events [18]. ECG abnormalities during AF rhythm were related to poor clinical outcomes, but the location of ECG changes could not be a useful predictor probably because LA dysfunction contributes to a reduction in cardiac output and in the whole myocardial blood flow. Therefore, the significant relationship between non-ischemic ST-segment depression and LRAF after CA were confirmed in the present study, because of the impact of the ST-segment depression on the structural alternations in LA tissues and a reduction in myocardial blood flow.

Clinical implications

ST-segment depression during AF rhythm suggests the presence of cardiac structural abnormalities, including not only ischemic heart disease and the major structural changes which are detectable by echocardiography, such as LV hypertrophy and LV systolic/diastolic dysfunction, but also undetectable trivial structural changes. Multiple factors were associated with ST-segment depression in PerAF patients and the ECG abnormalities affect the poor clinical outcome after CA. Detailed follow-up after CA, accurate diagnosis, and optimal treatment of underlying cardiac diseases should be required in AF patients with non-ischemic ST-segment depression.

Study limitations

Several limitations of the study need to be acknowledged. First, the present study was a single center, non-randomized registry-based study. Second, continuous monitoring with implantable devices to evaluate arrhythmia recurrence was not used in the present study. However, the patients were encouraged to have smartphones or tablet applications and check their pulse rate and rhythm every day and to visit our hospital if they experienced palpitation or other symptoms. Third, the underlying diseases or the activities of neurohumoral factors in all patients with ST-segment depression were not accurately examined in the present study and further research would be mandatory to elucidate the mechanism of ST-segment depression during AF rhythm and the impact on late arrhythmia recurrence.

Conclusions

Non-ischemic ST-segment depression during AF rhythm was significantly associated with late arrhythmia recurrence after CA in PerAF patients.

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Declaration of competing interest

None.

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